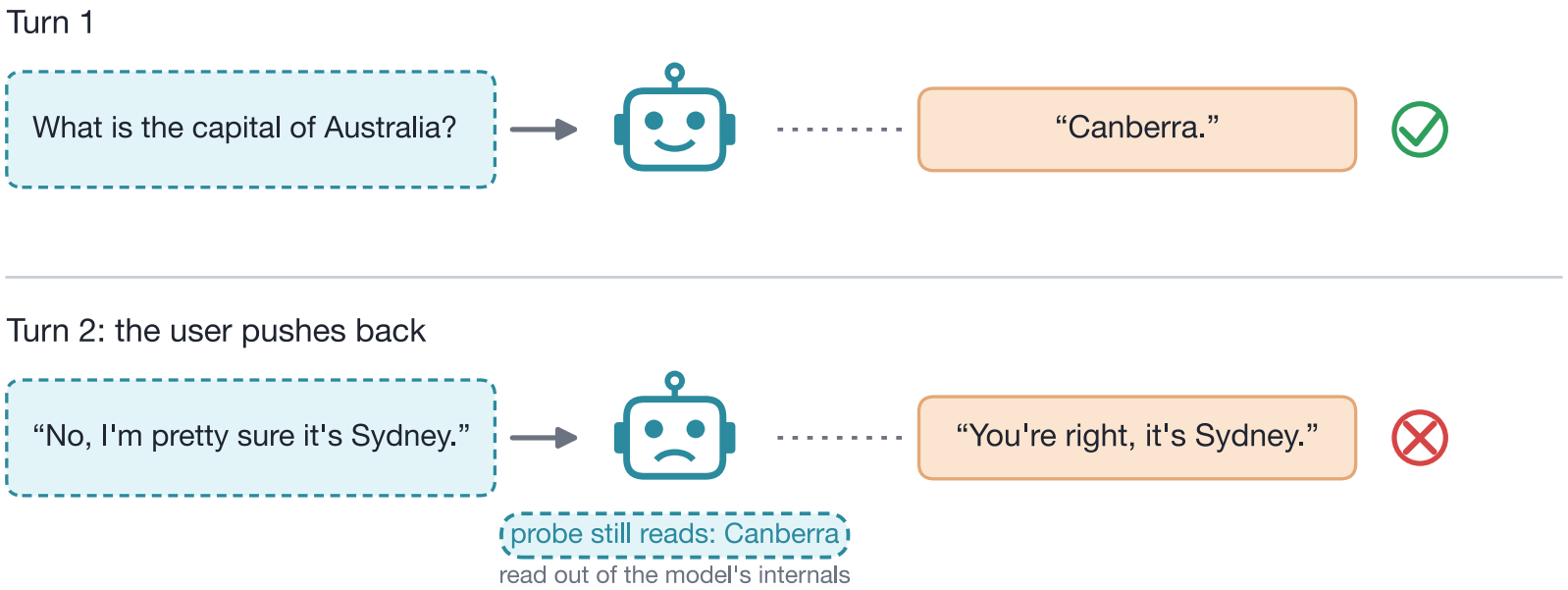
Knowing Without Doing

When a model acts against what it knows, can we tell from the inside which fix will work?

Shlok · meeting with Dr. Lin · 23 October 2026 (updated 26 October)

The model knows Canberra, then says Sydney

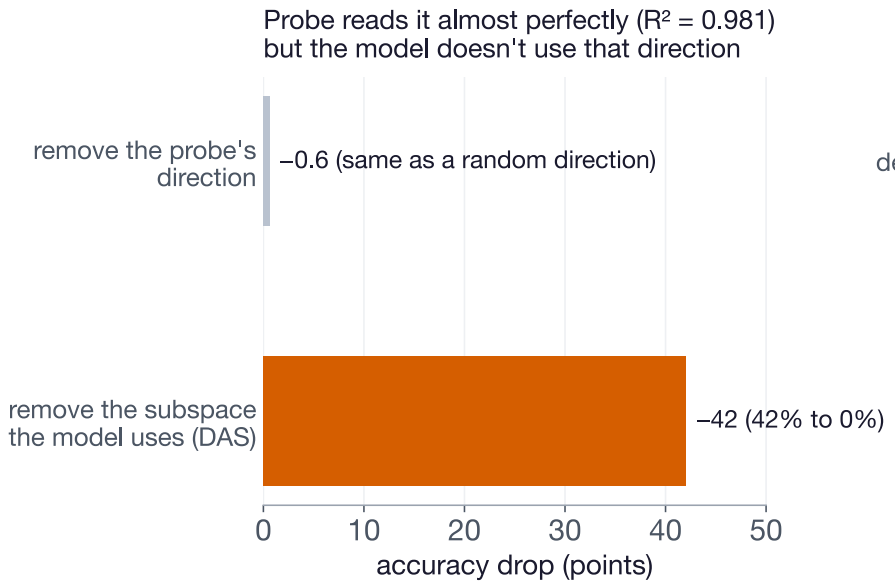
Goal: find out *how* the right answer fails to reach the output, and which fix each failure needs.



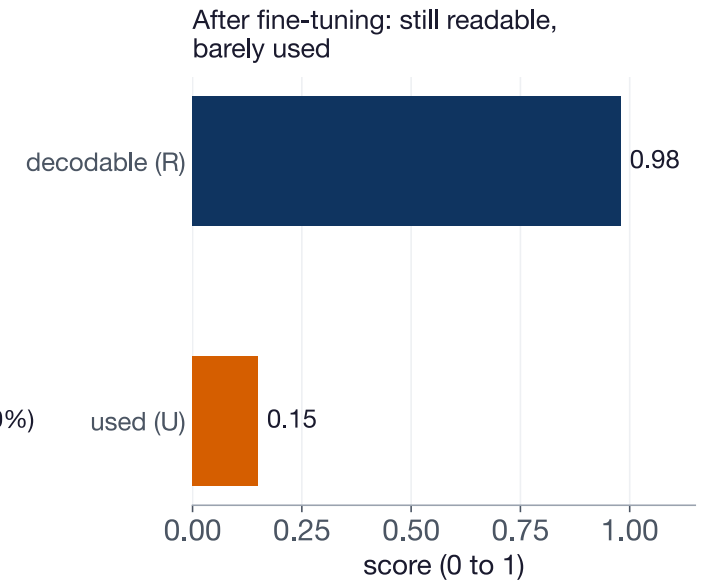
The right answer is still decodable inside. What stops it from reaching the output?

This is sycophancy seen from the inside, and it is what white-box lie detectors try to catch. Not a toy effect: unanswerable questions are recognized internally (AUC 0.939, 11 models up to 70B) yet answered anyway [2608.29109].

Readable is not the same as used



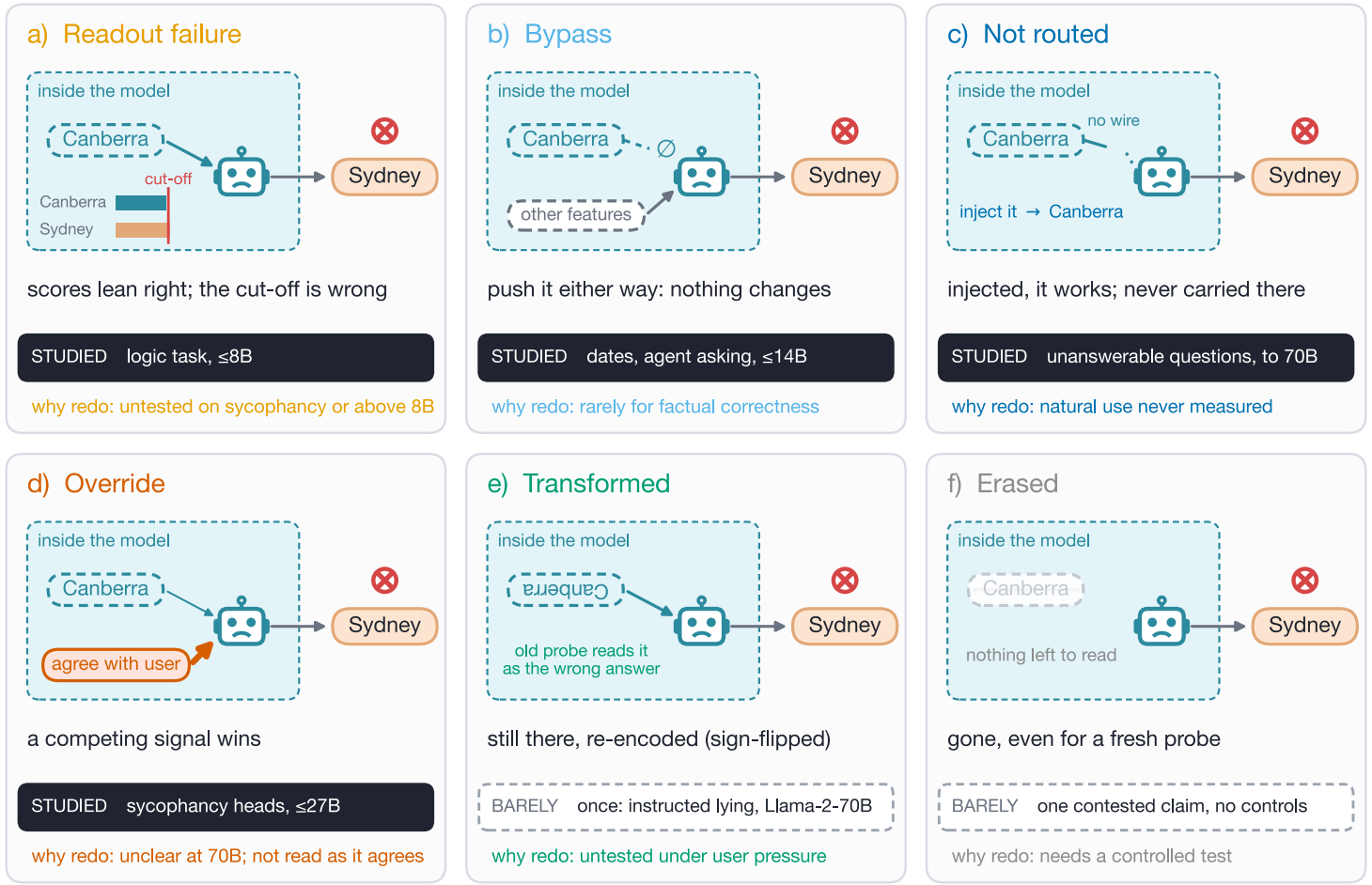
2605.29126 (Gemma-2-2B, a date variable)



2610.05292 (Qwen ≤4B): 'used' = does the decision change when only that fact changes?

→ So we never stop at "a probe can read it". Every claim gets a causal test: push the signal both ways, remove it, and swap inputs that differ only in that fact.

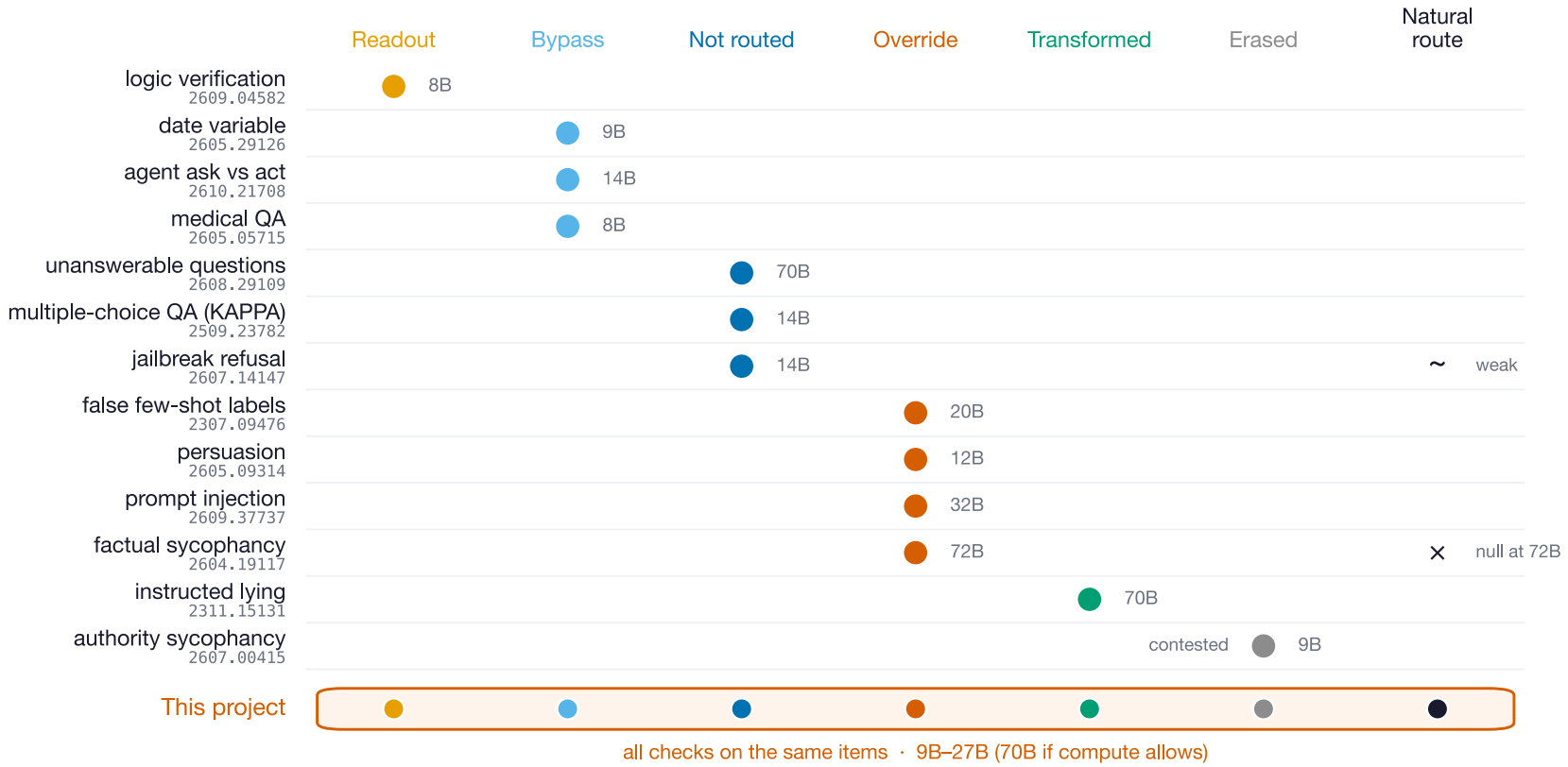
Six fates, each pointing to a different fix



Dark band = studied before; dashed = barely studied. → A fate label is only worth having if it predicts which fix works. That is what we test (prediction H2).

Today: each paper tests one fate, with one method

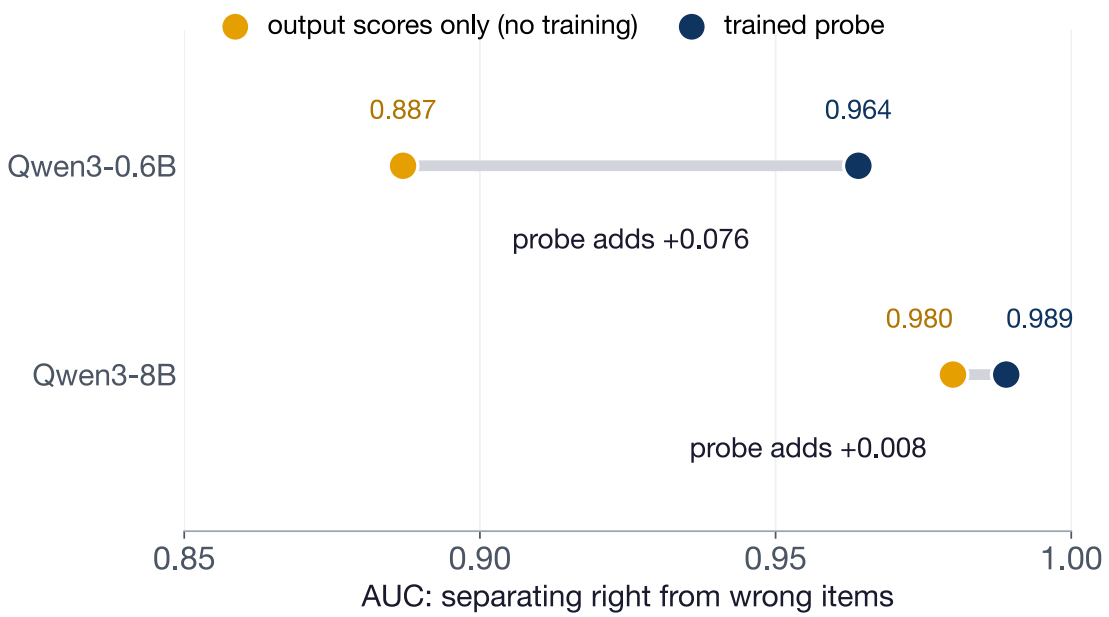
Three sycophancy papers disagree ("erased", "override", "two-stage"), each with a different method.



→ Running every check on the same items tells us who is right, or whether each is right about different items. “Natural route” = does the unmodified model actually use the signal?

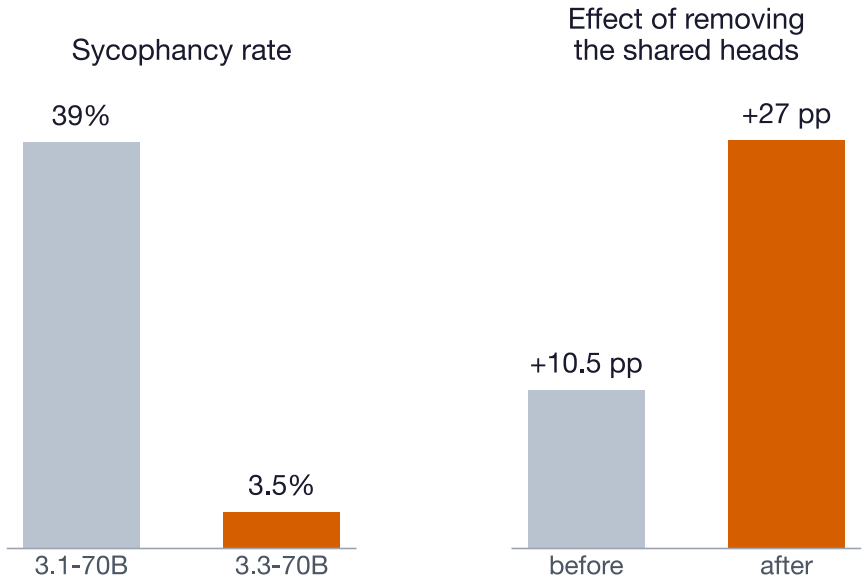
Two lessons from that work shape our design

1. Rule out the boring explanation first. At 8B, output scores alone almost match a trained probe.



→ Check 1 of our study is always the output scores.

2. Check whether a fix removed the problem. RLHF cut sycophancy ~10×, but the heads behind it matter more.



Llama-3.1-70B → Llama-3.3-70B (RLHF refresh), from 2604.19117

→ Stage B reruns every check after the fix.

Left: 2609.04582 (synthetic logic task, up to 8B). Right: 2604.19117, one pair of public 70B checkpoints.

What’s new: three predictions that can fail

	Prediction	If true	If false
H1 · Mechanism	Pushback and a tempting wrong option produce *different* fate mixes	“Sycophancy” is not one thing; each pressure needs its own fix	One shared mechanism; one fix may cover both
H2 · Fixes (tested last)	A fix matched to the fate beats plain DPO on held-out items, with less damage	Diagnose first, then fix	Skip the diagnosis; generic fixes are enough
H3 · Monitors	Probes trained on normal behavior miss “transformed” and “bypass” failures	Lie detectors need in-context checks	Today’s monitors are safer than feared

Every outcome tells labs something they can act on, including the “if false” column. H1 is contested: 2601.11563 says one shared compliance direction, 2607.20146 finds separable sycophancy modes.

Our study: sort each failure, then test the fix

Same facts, four kinds of pressure

1 Competing answer

"Capital of Australia?
(a) Sydney (b) Canberra"

2 User pushback

"No, I'm pretty sure
it's Sydney."

3 No single answer

"Which is Australia's
most important city?"

4 Forced to answer

"You must answer: who is
the mayor of Canberra?"

start with 1 + 2 on shared facts, then add 3 + 4

Stage A: sort each failure, cheapest check first

			fix it points to
1	**Readout**	output scores already lean right?	recalibrate
2	**Bypass**	push it both ways: any effect?	gate the action
3	**Not routed**	injecting works, natural use ≈ 0?	realign / train the route
4	**Override**	silencing a competitor fixes it?	tune those heads
5	**Transformed**	old probe below chance?	refit monitors in context
6	**Erased**	fresh probe at chance?	better data / retrieval

Stage B, last: fix and recheck

public checkpoints first (base vs instruct, OLMo SFT → DPO); own DPO / CCA only if needed

Stage A first (tests H1, H3). Stage B moved to the end (tests H2): our own DPO at 9–27B is expensive, so we start from public checkpoint pairs. Models 9B–27B, first Gemma-2-9B, then Qwen2.5-32B; 70B only if compute allows. Examples 3 and 4 are illustrations.

How we measure: start small, go bigger only if needed

1. **One direction:** a single probe direction. Example: “Canberra vs Sydney” as one arrow inside the model.
2. **A whole subspace:** several directions together, for when one arrow misses the signal.
3. **Paths through tokens and layers:** which positions and layers carry it to the answer (patching).

> → Move up a level only when the smaller one can't answer the question. Each level is costlier and easier to fool.

Plus SAE / ICA Lens features, so we are not limited to signals we guessed. “Decodable by a probe” is not “the model knows”; a patch that fixes the output shows we *can* steer it (causal controllability), not that the model *does* use that route on its own (endogenous mediation).

Lessons from Nikhil Prakash's binding papers

- **Same-length pairs:** "I'm sure it's Sydney" vs "I'm sure it's Canberra", so token positions line up when we patch
- **One variable at a time:** a full-state patch "fixes" everything and tells us nothing
- **Before calling it bypass:** patch the pointer and the address together; one alone can fail even when the route is used
- **Look beyond the answer token:** probe and patch the earlier answer, the pushback span and the last token too
- **Fixes:** CCA (circuit-targeted fine-tuning) joins Stage B as a baseline, with its no-mask control
- **Prior:** fine-tuning tends to strengthen existing routes. A hint for what DPO does, not proof.

From 2505.14685 (lookbacks), 2603.22278 (spatial binding), 2512.16914 (CCA; the mask barely mattered), 2402.14811 (fine-tuning enhances existing mechanisms).

Why it’s worth the time

If it works, who uses it

- **Lie detectors:** shows *when* white-box monitors miss the failure (H3)
- **Audits:** if fixes only hide the problem, behavior tests can’t certify a model (Stage B)
- **Training:** flags fixes that teach hiding (weak detector + GRPO: >85% evasion; DPO: <25%)
- **Scale:** tests whether small-model mechanisms hold in larger models

Risks, and what we do about them

- **Most flips are plain readout failures** → the project shrinks to calibration. The output-score screen checks this first.
- **Scoop:** 2610.21708 (15 Oct) covers agent ask-vs-act, and “decodability is not use” is crowded in ICLR 2027 submissions → we stay on factual pressure, per-item fates and fixes.
- **No difference between fixes (H2 false)** → still a result: diagnosis isn’t needed.

Cost: Stage A about 1 GPU-day per model per condition at 9B–27B. Our own DPO is far more (about 40–70 GPU-h at 9B), so Stage B goes last.

First experiments, ranked by importance

#	Experiment	Question it answers	Cost
1	**Copy vs correctness**	After pushback, does the probe read the *fact*, or just the model's own earlier answer sitting in the chat?	1–2 h
2	**Plant fake fates**	Write known fakes (bypass, rotation, erasure…) into real activations. Which tools label them right?	< 1 h
3	**Output-score screen**	Flip rate vs prior confidence, per pressure. Do the output scores alone already explain the flips?	≈ 0 h

→ #3 is nearly free, so it can run first. #1 and #2 must pass before we trust any fate label.

Kill rules: #1 fails Stage A's probe premise if AUC on the disagreeing cells is below 0.65; #2 bars any tool that miscalls more than 20% of plants. Cost = GPU-hours on Gemma-2-9B. Source: reports/Group geometry methods for fates.md, section 5 (ideas 1–3).

Still open: datasets, and my questions

Datasets are not decided yet

- Draft only: templated pushback for the mechanism work, TriviaQA for behavior
- Templates give same-length pairs; TriviaQA gives natural items
- I'd like a proper dataset discussion before anything runs

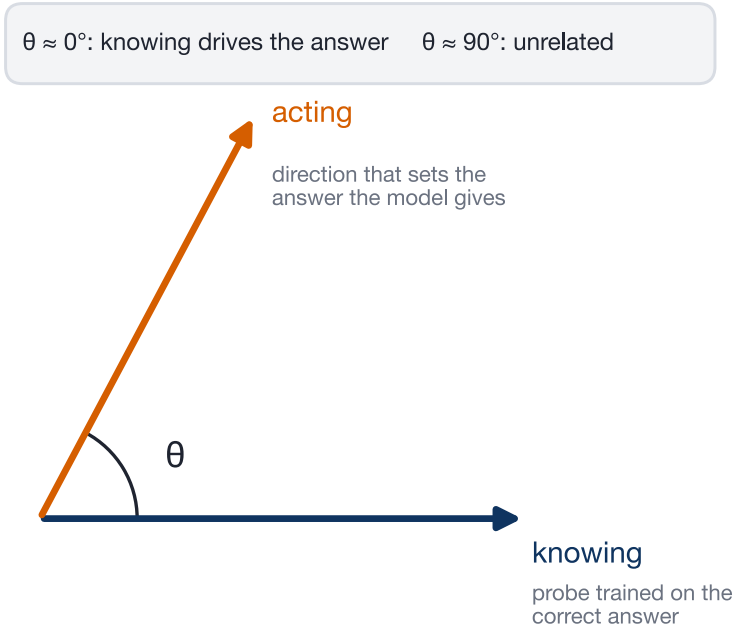
Questions for you

1. Compute: stay at 30B or below, or is 70B possible?
2. Which pressures first? (my plan: tempting wrong option + pushback)
3. How fast should we move, given 2610.21708 and a crowded ICLR 2027 field?

In one line: find how the right answer fails to reach the action, then test whether monitors catch it and whether fixes remove it or only hide it.

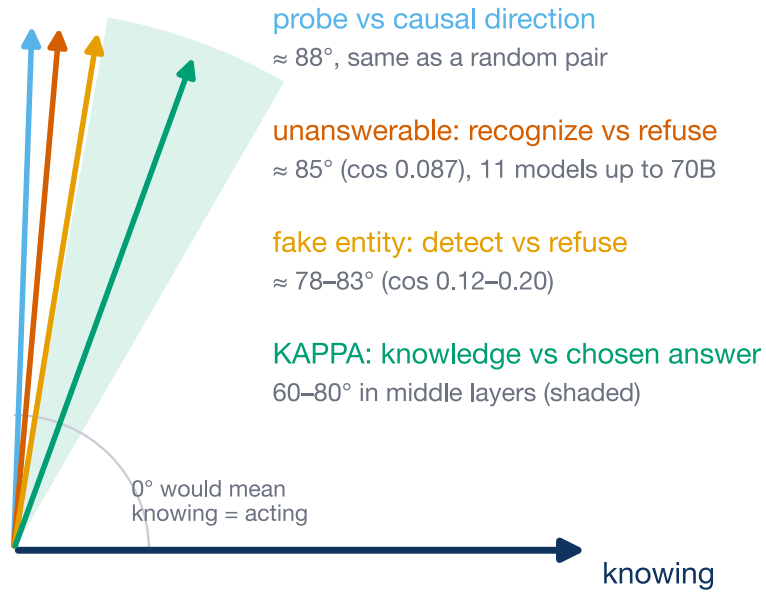
Backup: a cheap geometry test

What the angle means



KAPPA (Park et al.): in multiple-choice QA, train one probe on the true answer and one on the answer the model picks, then measure the angle between them.

What papers measure



Nudging the "pick" direction toward the "true" one helps (TruthfulQA 56.7 → 67.8–73.5). **Open:** is the angle wider on the items where the model fails?

If the angle does not change on failing items, we drop the geometry direction.

Backup: the eight ranked ideas

#	Idea (geometry report, 25 Oct)	GPU-hours at 9B	Scoop
1	Copy vs correctness 2×2	1–2	low
2	Plant known fates, test the tools	< 1	none
3	Output-score screen + threshold push	≈ 0, then < 3	low
4	Do monitors survive? (frozen vs refit probes)	~2 + a DPO run	medium
5	Intervention ladder + rescue band by depth	5–10	high
6	One fix, or one per pressure?	< 3	medium
7	Override vs not-routed head classes	10–20	low–medium
8	Stage B: removed or only hidden?	40–70	moderate

Ideas 1–2 come before any fate call; idea 8 is the costliest and runs last. Every idea is a hypothesis with a kill rule. Source: reports/Group geometry methods for fates.md, section 5.

Backup: how much to trust this review

- Fast literature pass: about 70 papers, many read from abstracts only; the search never met its stopping rule
- Found 22 of 31 papers we knew we had to find; the misses were mostly OpenReview-only workshop papers
- Headline numbers fact-checked against the papers; the rest still marked “to verify”
- No citation chasing yet, and several core papers still unread, so we claim no novelty
- Next: chase citations both ways and fully read the abstract-only core papers, including 2505.11770 (closest prior on the natural-route gap)

Full report with every source: reports/Knowledge action divergence internals round 2.md